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Discrete Mathematics

10Week [13 April - 19 April]

Quiz 10

Started on	2024-04-24 21:13
State	Finished
Completed on	2024-04-25 08:27
Time taken	11 hours 14 mins
Grade	10.00 out of 10.00 (100%)

Question 1

Complete

Mark 3.00 out of 3.00

Let $a = (1010)_2$ and $b = (1101)_2$.

Find ab using the formula

$$ab = (2^{2n} + 2^n)A_1B_1 + 2^n(A_1 - A_0)(B_0 - B_1) + (2^n + 1)A_0B_0,$$

where $a = 2^n A_1 + A_0$, $b = 2^n B_1 + B_0$, $n = 2$ and

$$A_1 = (a_{2n-1} \dots a_n)_2, \quad A_0 = (a_{n-1} \dots a_0)_2$$

$$B_1 = (b_{2n-1} \dots b_n)_2, \quad B_0 = (b_{n-1} \dots b_0)_2.$$

Compare your answer with $ab = (10000010)_2$.

Attach your solution below.



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Comment:

Question 2

Correct

Mark 1.00 out of 1.00

How many rabbits (not pairs) you will have at the end of the first year in the Fibonacci's Rabbits model?

Select one:

- ☒ 288 ✓
- ☐ 144
- ☐ 89
- ☐ 178
- ☐ 233

Your answer is correct.

The correct answer is: 288

Question 3

Correct

Mark 1.00 out of 1.00

Choose a linear homogeneous recurrence relation(s) with constant coefficients.

Select one or more:

- ☒ $a_n = 5a_{n-1} + 3a_{n-2} - 8a_{n-3} + 40a_{n-4}$ ✓
- ☐ $b_n = \sin(b_{n-1}) + 5b_{n-2}$
- ☐ $T_n = T_{n-3} + (2n + 3)T_{n-6}$
- ☒ $P_n + P_{n-2} + P_{n-4} = 0$ ✓
- ☐ $Q_n = Q_{n-3} + 48$

Your answer is correct.

The correct answers are:

$$a_n = 5a_{n-1} + 3a_{n-2} - 8a_{n-3} + 40a_{n-4}$$

$$P_n + P_{n-2} + P_{n-4} = 0$$

Question 4

Complete

Mark 3.00 out of 3.00

Solve the recurrence relation $a_n = a_{n-1} + 6a_{n-2}$ with the initial conditions $a_0 = 3$ and $a_1 = 6$.

Attach your solution below.

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Comment:

Question 5

Complete

Mark 2.00 out of 2.00

What form does a particular solution of the linear nonhomogeneous recurrence relation

$$a_n = a_{n-1} + 6a_{n-2} + F(n)$$

have when

$$F(n) = (n^2 - 5)3^n \text{ and } F(n) = (n - 5)7^n ?$$

Attach your solution below.

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Comment: